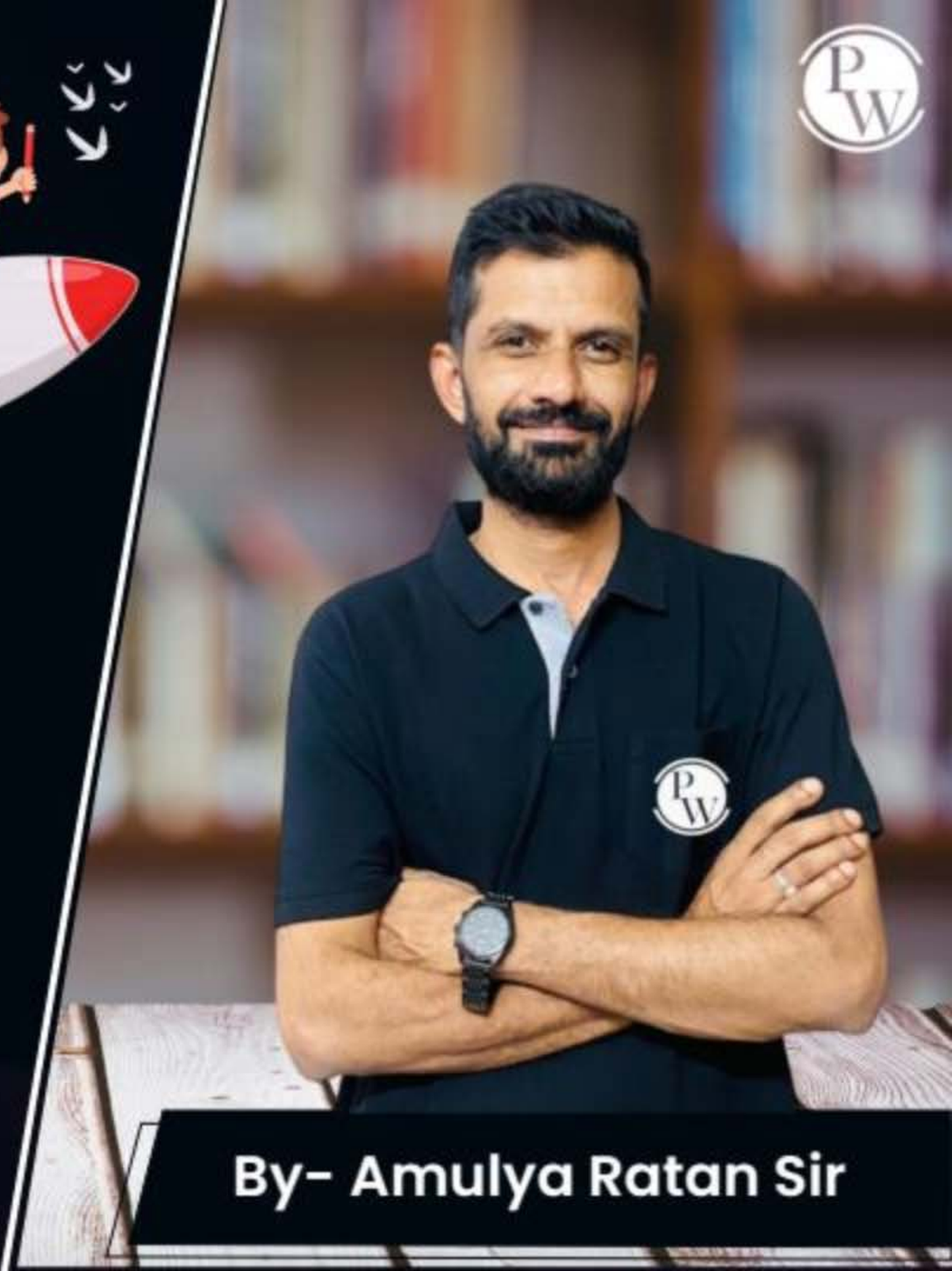
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 11



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Counting Theory

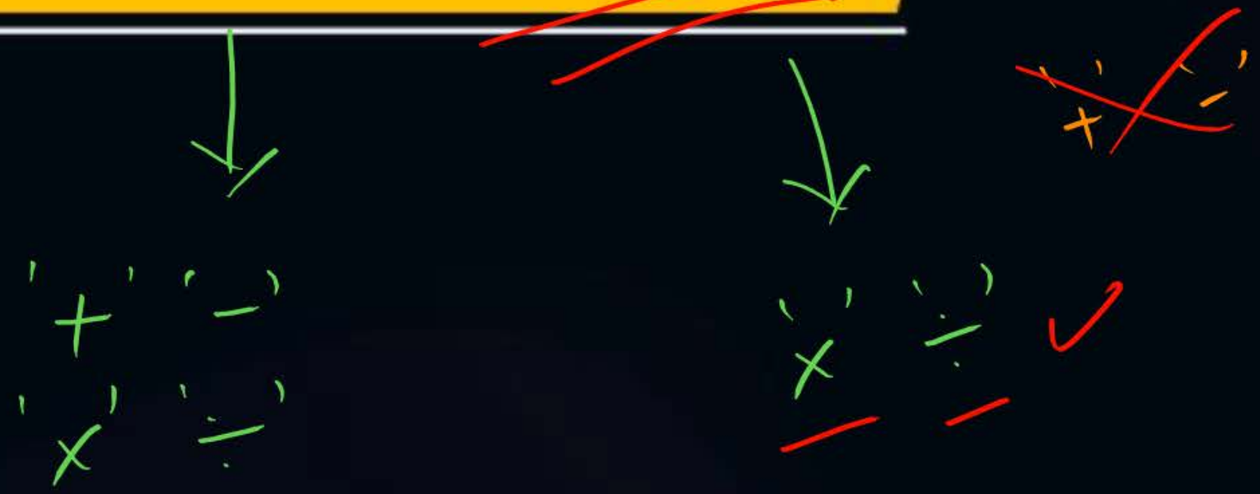
Topics to be Covered



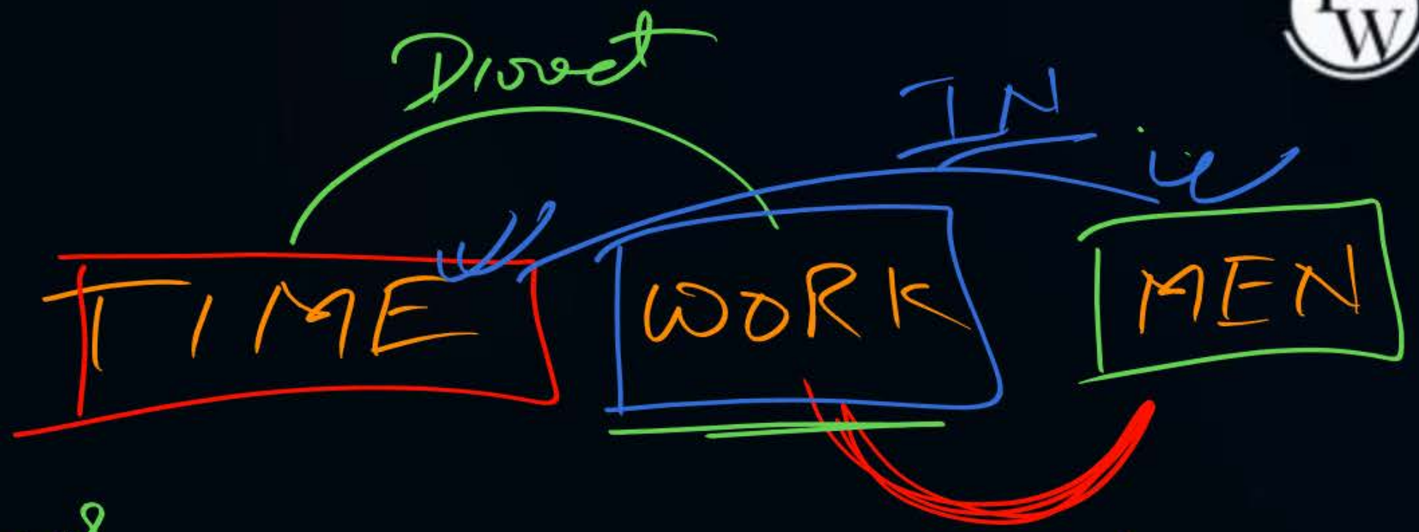
Topic-1 Time & Work

Topic-2 Chain Rule

Direct & Inverse:



Time & Work



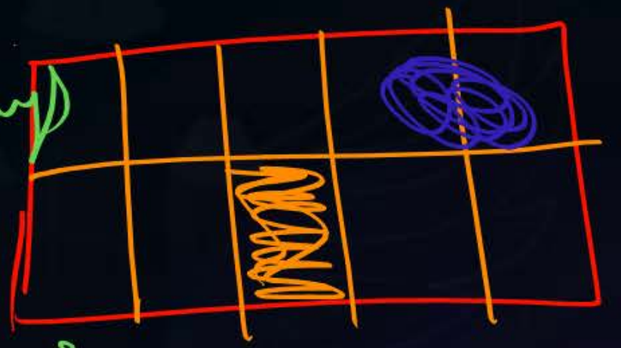
$\frac{1}{10}$ P = $\frac{10}{1}$ days

$\frac{1}{20}$ Q = 20 days

P & Q = $\frac{1}{10} + \frac{1}{20} = \frac{3}{20} \Rightarrow \frac{20}{3} = 6.\overline{6}$ days

OR

$6\frac{2}{3}$ days



Let's Try:

$A = 20$ ✓ | $B = 30$ ✓

$\frac{1}{20} + \frac{1}{30} = \frac{3+2}{60} = \frac{5}{60} \rightarrow \frac{60}{5} = 12$

Together

$A = \frac{1}{25}$ $B = \frac{1}{40}$

$\frac{1}{25} + \frac{1}{40} = \frac{8+5}{200} = \frac{13}{200}$

$A = 25$ | $B = 40$ $\frac{200}{13} \rightarrow 15\frac{5}{13} = 15.38$ days

P+Q

$P \& Q \Rightarrow 40$ | $P = 60$ | $Q = ?$

$\frac{1}{40} - \frac{1}{60} = \frac{1}{120}$

120 days

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the total work would be completed ?

$\frac{1}{10} + \frac{1}{20} = \frac{3}{20}$ ×6 → 2 days ×6

1	2	3	4	5
A	B	A	B	A ...

$\frac{2}{20}$? $\frac{1}{10}$?

$\frac{18}{20}$ → 12 days

13 days

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'B', 2nd day 'A', 3rd day 'B'.....so on. Then in how many days the total work would be completed ?

13th (B)

$\frac{1}{10} - \frac{1}{20} = \frac{1}{20}$

1 | 2 | 3 | 4 | 5
B | A | B | A | B

$\frac{1}{20} + \frac{1}{10} = \frac{3}{20} \xrightarrow{\times 6} 2$ days

$\frac{18}{20} \Rightarrow 12$ days

1 male $A = 10$

$\frac{1}{10}$ :: $A = 10 \times \frac{1}{20}$

$\frac{x}{20} = \frac{1}{10}$?

$x = 10$

$= \frac{1}{2}$

$13\frac{1}{2}$ OR 13.5 days

[MCQ]

#Q. A can do a work in 25 days whereas B in 40 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the work would be completed?

A. $15\ ^{5}/_{8}$ days

B. $30\ ^{5}/_{8}$ days

C. 31

D. 16

$30\frac{5}{8}$ OR 30.625

$A = \frac{1}{25}$ | $B = \frac{1}{40}$

1	2	3	4	5
A	B	A	B	A

$\frac{1}{25} + \frac{1}{40} = \frac{13}{200}$ (×15) $\Rightarrow$ 2 days (×15)

1	2	3	4
B	A	B	A

$\frac{8}{200} = \frac{1}{40}$?

$\frac{195}{200} \Rightarrow$ 30 days

31 days

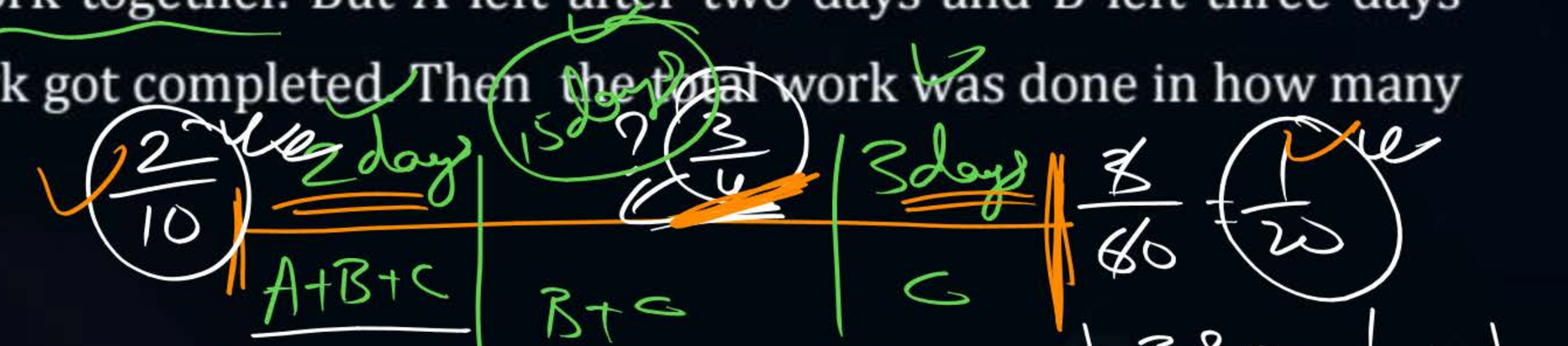
1 work A = 25

$\frac{1}{40}$ :: $A = 25 \times \frac{1}{40} = \frac{5}{8}$

[MCQ]

$A = \frac{1}{20}$ u | $B = \frac{1}{30}$ u | $C = \frac{1}{60}$ u

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?



20 days

$A+B+C = \frac{1}{20}+\frac{1}{30}+\frac{1}{60} = \frac{6}{60} = \boxed{\frac{1}{10}}$

$\frac{2}{10}+\frac{1}{20} = \frac{8}{20} = \frac{1}{4}\left(1-\frac{1}{4}=\frac{3}{4}\right)$

$B\&C = \frac{1}{30}+\frac{1}{60}$

$= \frac{3}{60} = \frac{1}{20}$

$\frac{3}{4}$ work B&C = 20

1 work B&C = $\frac{20}{3}$

B&C = $\frac{1}{20}+\frac{3}{4}$

= 15 days

[MCQ]

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

'x' days

$$\frac{x}{60}+\frac{x-3}{30}+\frac{2}{20}=1$$

$$x+2x-6+6=60$$

$$3x=60$$

$$x=20$$

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$ | $C = \frac{1}{60}$

#Q. A can do a work in 10 days whereas B in 20 days and C in 60 days. They started the work together. But A left at the end of 3rd day and B left at the end of 5th day. Then the remaining work was done by C in how many days?

3 days	2 days	?
A+B+C	B+C	C

'x' days

A. 27

B. 35

C. 31

D. 22

$\therefore x = 27$

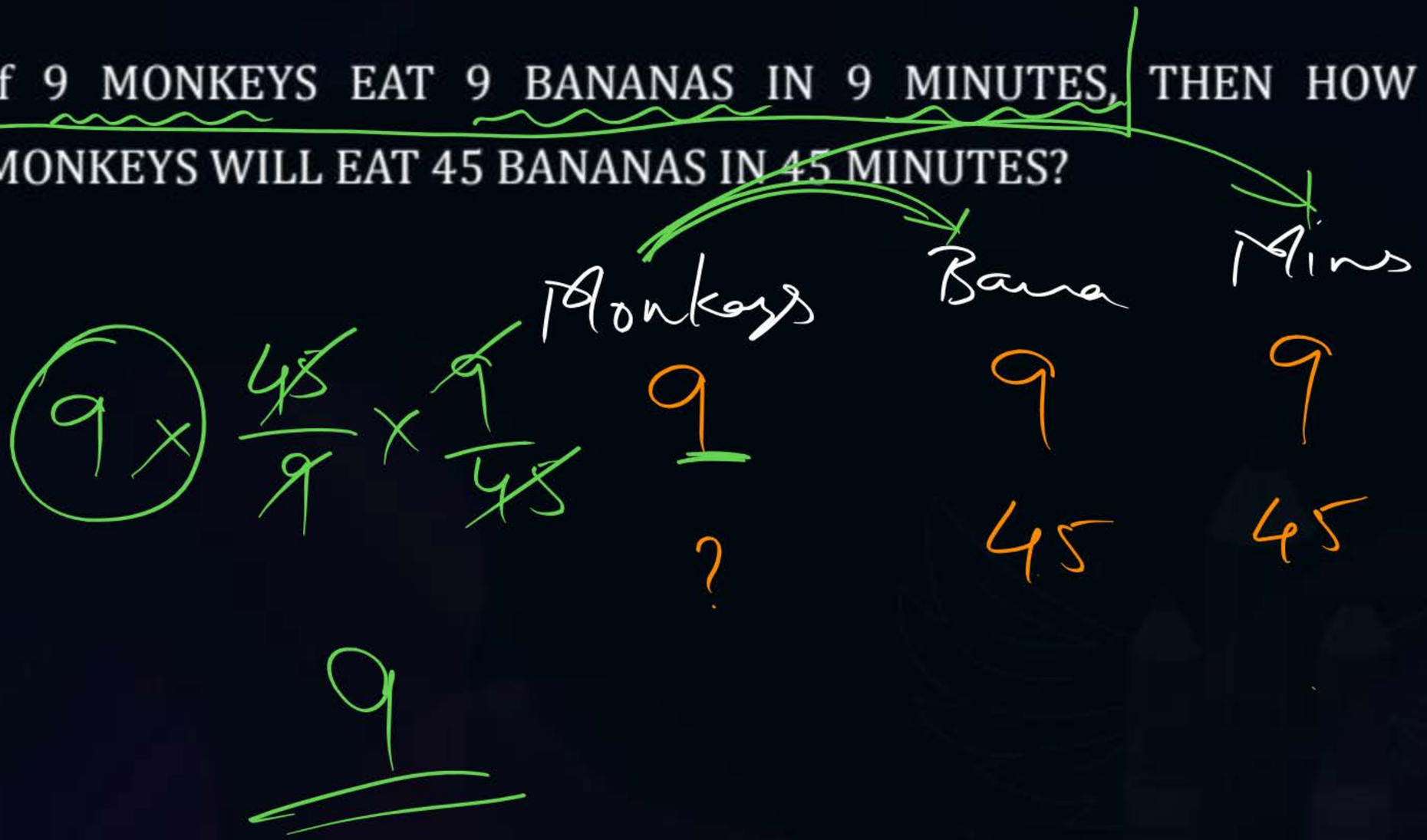
$27 - 5 = 22$ days

$\frac{x}{60} + \frac{5}{20} + \frac{3}{10} = 1$

$\Rightarrow x + 15 + 18 = 60$

[MCQ]

#Q. If 9 MONKEYS EAT 9 BANANAS IN 9 MINUTES, THEN HOW MANY MONKEYS WILL EAT 45 BANANAS IN 45 MINUTES?



Chain Rule:

↑ ① $\times \frac{3}{2}$

↓ ① $\times \frac{2}{3}$

%	No.
32%	78
65%	?

$78 \times \frac{65}{32}$

32% of x = 78

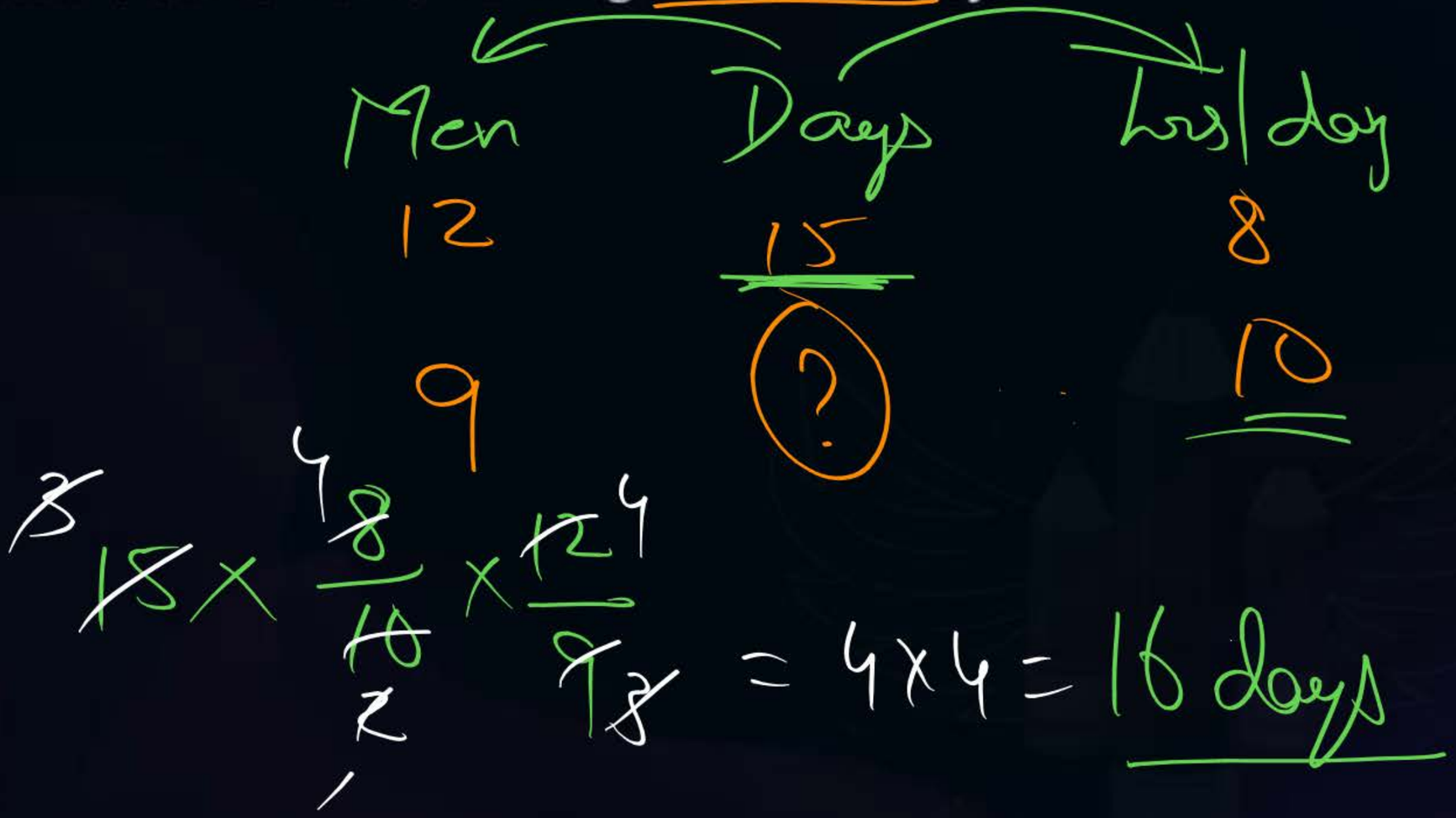
$\frac{32}{100} \times x = 78$

$x = 78 \times \frac{100}{32}$

65% of $x = \frac{65}{100} \times 78 \times \frac{100}{32}$

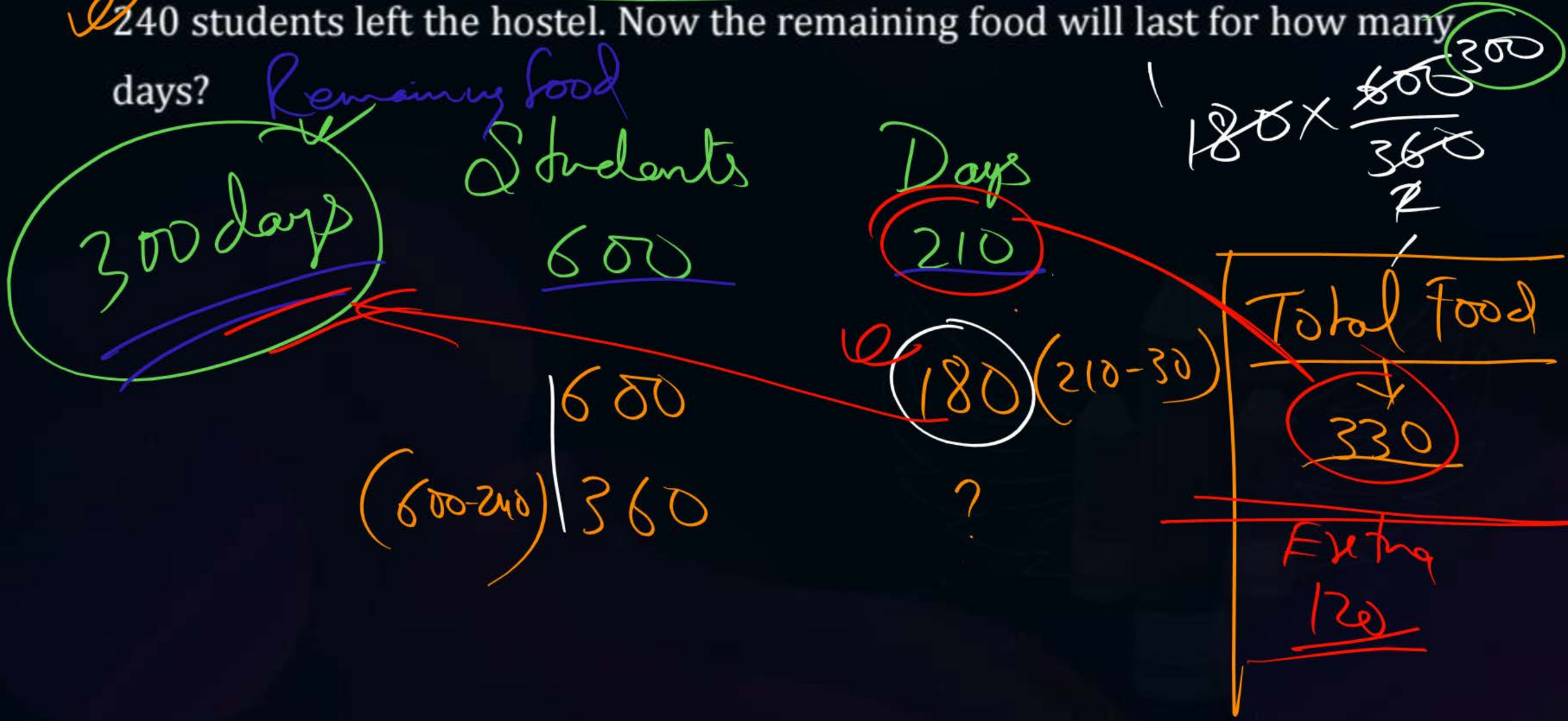
[MCQ]

#Q. 12 men can do a work in 15 days working 8 hours a day. In how many days can 9 men do the same work, working 10 hours a day?



[MCQ]

#Q. A hostel with 600 students had sufficient food for 210 days. After 30 days, 240 students left the hostel. Now the remaining food will last for how many days?



[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If that clock represents 1 pm the same day, what must be the correct time?

Assignment

[MCQ]

#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

Assignment

A. 75

B. 100

C. 50

D. None of these

[MCQ]

#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

Assignment

A ₹ 8000

B ₹ 6000

C ₹ 1800

D ₹ 2040



2 mins Summary

Topic Time and Work

→ chain Rule

THANK - YOU